

SWON Fun test

1) Explain types of medium?

- * UTP cable (electrical signal)
- * Radio frequency (wireless)
- * Fiber optics (Light signal)

2) Types of network communication?

- * unicast (one to one)
- * Multicast (one to group)
- * Broadcast (one to All)

3) write down pc and switch 8 pins in order?

PC	switch
1 +T	1 R
2 -T	2 R
3 +R	3 T
4 +V	4 V
5 -V	5 V
6 -R	6 T
7 x	7 x
8 x	8 x

This is straight cable.

4) color code of T568B

pin wire color

1 orange

2 orange

3 Green

4 Blue

5 Blue

6 Green

7 Brown

8 Brown

This is used for straight-through Ethernet patch cables today

5) pc-pc cable crimping type?

pc-pc is cross cable
Router-switch is straight cable.

6) Auto-Mdix function?

Automedix is a feature that automatically detects whether a straight-through or crossover Ethernet cable is required and configures the interface accordingly without manual configuration.

7) function of wired and wireless nic

NIC (Network Interface Card) is a hardware component that allows device to connect a network.

wired nic connects a computer to a network using ethernet cable.

wireless nic connects a device to a network using radio signals.

8) PoE?

PoE is power over Ethernet that allow a network Ethernet cable to carry both data and electrical power to device.

9) PoE switch vs PoE injector

A PoE switch provides both network connection and power to multiple PoE devices, whereas a PoE injector add power to an ethernet connection, usually a single PoE device, when the existing switch does not support PoE.

10) IP parameter of 181.0.20.0

network ID: 181.0.0.0

First IP: 181.0.0.1

Last IP: 181.255.255.254

Broadcast IP: 181.255.255.255

Total no. of IP: 65,435

previous IP: 181.0.19.255

next IP: 181.0.20.1

11) why subnet mask

Subnet mask is used to separate the network portion and host portion of an IP address and to determine which device belong to same network

EX: 181.0.20.0

network host

12) RDP?

(Remote Desktop protocol) RDP is a Microsoft protocol used to remotely access and control a Windows computer (server) over a network

RDP port no: 3389, its default TCP.

13) netbios?

Netbios can help identify server-PC by its network name and establish communication with it.

It's used in windows network. ports are UDP 137, UDP 138, TCP 139.

14]

Explain OSI layer when ping 10.0.0.10 to 10.0.0.20

- * The application layer: generate a echo data, with encapsulation help of ICMP

- * presentation layer: encapsulate data echo with adding format.

- * session layer: encapsulate echo data format adding with time.

- * Transport layer: encapsulate echo data format, time with port no with help of tcp/udp protocol.

- * network layer: encapsulate echo data format, time port no with source IP and destination IP is called packet encapsulation.

- * Data link layer: encapsulated echo data format, time, port no, sip of dip now with source mac of destination mac with help of ethernet protocol. Is called frame encapsulation.

- * physical layer: transmit that data through utp data to the connected to pc, the connect pc will receive the data.

15]

when we need switch device?

- * switch when we have multiple devices that need to communicate on the same LAN

- * EX: All ip network in 192.168.1.0/24 switch can connect them.

- * switch mainly work at layer 2 and use mac address to forward ethernet frame.

16) When do we need Router device?

* Use a router when communication is required b/w different networks.

EX: PC1: 192.168.1.10, PC2: 192.168.2.10

* Router works at Layer 3 and it has address / routing tables to forward packets.

17) Switch packet flow?

When a frame enters a switch, the switch learns the source MAC address and incoming port. It then checks the destination MAC in its MAC address table. If the destination is known, it forwards the frame through the corresponding port, or else, Broadcast to all ports in switch to find destination MAC.

18) Router packet flow?

A router forwards packets b/w diff IP nets using its routing table.

When a packet reaches a router, the router will check the destination IP, performs a routing-table look up, selects the best route and outgoing interface.

It will check network ID, netmask, default gateway. It will be correct forward the packet, not correct drop the packet.

19) What is directly connected networks?

A directly connected network is a network that is reachable through a router that interface that is physically connected and operational. The router automatically adds it to its routing table.

20) What is command to enable telnet?

Line vty 0 4

login local

transport input telnet

21)

How to set console port authentication?

line console 0

login local

exit

(username / password)

22)

write system route table of 129.0.0.0

network	netmask	Gateway
129.0.0.0	255.255.0.0	129.0.129.0
129.0.129.0	255.255.255.0	129.0.0.1

23)

Explain destination host unreachable?

It don't have route in system route table so it occur.

24)

Explain request timed out?

It won't receive any reply from target device

25)

Explain importance of default gateway?

The default gateway is the ip address of router interface used by host to forward traffic to destination outside its local network. without a correct default gateway, a device can usually communicate with its own subnet but can't reach remote network.

26) static route?

A static route is a manually configured route that tells a router the destination network and the next hop or exit interface to use to reach that network.

27) command syntax of static route?

`ip route <destination-network> <subnet-mask>
<next-hop-ip>`

EX: `ip route 192.168.2.0 255.255.255.0 10.0.0.2`

28) when load sharing will happen in Cisco router?

Load sharing occurs when a router has multiple equal-cost paths to same destination. The router can install multiple next hops and distribute traffic across those paths using EIGRP (equal cost load sharing).

29) Type of load sharing

same path

- * per-destination $\rightarrow$ particular traffic sent through same path
- * per-packet $\rightarrow$ Individual packets are distributed across diff available paths.

30) Layer 3 loop in deep?

A Layer 3 loop is a routing condition in which a packet is repeatedly forwarded between routers because their routing tables incorrectly point toward each other for same destination, then each router decreases the IPv4 TTL by one, and when TTL reaches zero, the packet is discarded and an ICMP Time Exceeded message is normally generated.

31) TTL and its Importance?

Time to live (TTL) is an IP header field used to prevent packet from circulating indefinitely. Each router decrement the TTL by one, and when it reaches zero, the packet is discarded and an ICMP time exceeded message is normally sent to the source.

32) ARP in layer 2 & layer 3?

- * ARP resolves IP address $\rightarrow$ MAC address
- * ARP Request uses layer 2 broadcast.
- * ARP Reply is normally unicast.
- * A layer 2 switch forwards ARP frames based on MAC address.
- * ARP operates only with the local layer 2 broadcast domain.

Layer 3 ARP is not layer 3 protocol. It supports layer 3 IP communication by resolving the IP address of next hop device into its layer 2 MAC address. The host ARP's for the default gateway, not for remote destination host.

33) What is MTU?

Maximum Transmission Unit (MTU) is maximum size of an IP packet that can be transmitted over a network interface without fragmentation. The standard ethernet MTU is 1500 bytes.

34) What is segment, syn no, window size?

Segment is the unit of data created by TCP at the Transport layer (layer 4).

The sequence number identifies the position of data in the TCP stream

EX: seq no: 1001, 1002
window size identifies how much data the receiver is willing to accept, providing TCP flow control

35)

Importance of three way handshake?

TCP-3 way handshake is important because it establishes a reliable connection, synchronizes synchronous numbers, confirms both sides are ready, and negotiates TCP parameters before data transmission begins.

36)

diff b/w linux and windows.

Linux is an open-source operating system that is highly customizable and widely used for servers and networking.

Windows is a proprietary OS widely used for desktops, business applications and micro-soft-based environments.

37)

Kernel?

The kernel is core component of an operating system that manages hardware resources such as CPU, memory, storage and network devices, and provides services to application.

38)

diff b/w client OS and server OS?

Client OS is designed mainly for end-user activities, while server OS is designed to provide network services and resources to multiple clients reliably often 24/7.

39) web services?

web services is a software service that enable different application to communicate and exchange data over a network using standard such as HTTP/HTTPS commonly through REST API or SOAP.

40) Importance of DNS?

DNS is important because it translates human-readable domain names into IP address, allowing clients to locate servers and access resources easily.

41) FQDN

fully qualified domain name (FQDN) it's the complete DNS name of a host, including the hostname and domain.

EX: www.google.com
↓ ↓
zone hostname host

42) packet flow of lab.surretnetwork.com?

The client first performs DNS resolution to obtain the server's IP address. If the server is remote to client uses ARP to find the default gateway's MAC address, sends the IP packet to the router, and the packet is routed through the ISP and Internet to the destination server. For HTTPS, TCP connection establishment and TLS negotiation occur before the encrypted HTTP request and response.

43) what is port number?

A port number is a 16-bit number (0-65535) used by TCP or UDP to identify a particular application or service on a device. for ex HTTP's port 80, SSH 22, Telnet 23.

44)	SNTP	→ network Time protocol	123
	IMAP	Internet Message Access protocol	143
	IMAPS	IMAP over SSL/TLS	993
	POP3	Post office protocol v3	110
	POP3S	POP3 over SSL/TLS	995

45) DHCP and DHCP process?

DHCP is protocol that automatically assigns IP address and other network parameters to client. DHCP server is responsible for providing this configuration. The DHCP process: DORA process

* ~~Discover~~ * DISCOVER

* OFFER

* REQUEST

* ACKNOWLEDGMENT

46) How DHCP ^{works} process on same & diff network?

DHCP client and server are on the same network, the DHCP ~~discover~~ broadcast can reach server directly. when different network, router doesn't normally forward the broadcast, so configure DHCP Relay to configure DHCP process to function in diff network.

47) command to enable dhcp relay agent
ip helper-address